

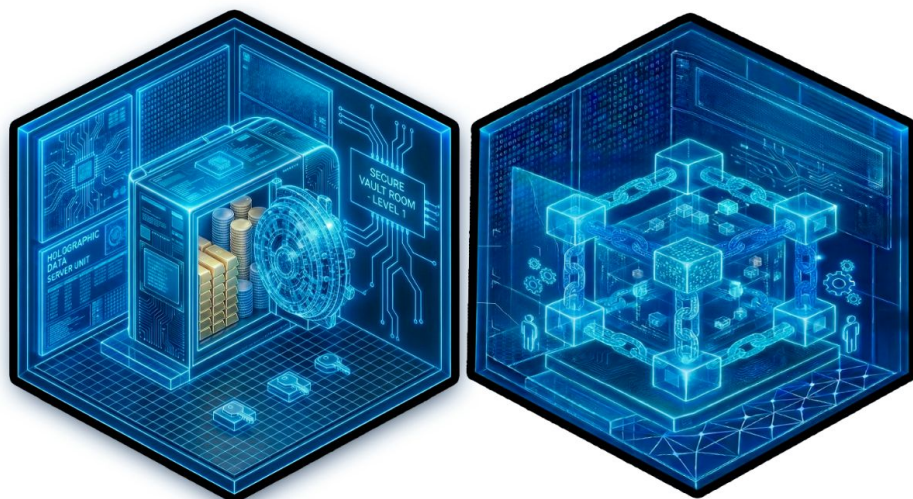


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TOKENIZATION OF DEPOSITS

BANKING TOWARDS INSTANT SETTLEMENT WORLD WIDE

**TOKENISED
DEPOSITS**



Abstract

*The **tokenization of commercial bank deposits** represents a structural **transformation** in how **regulated money is represented, transferred, and settled**. Unlike **stablecoins**, which are **liabilities of private issuers**, **tokenized deposits** are **direct liabilities of regulated commercial banks**, carrying the full weight of deposit insurance, capital requirements, and AML/KYC compliance. This paper examines the **architecture, global institutional deployments, and regulatory landscape** of tokenized deposits as they stand in **2026**, with particular attention to production-grade systems across the United States, Europe, Asia-Pacific, and the Middle East. The **central finding** is that **tokenized deposits are no longer theoretical**: they are live, processing real transactions, and embedding themselves into the settlement layer of core financial infrastructure.*

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1. Introduction

The **financial system** is undergoing its most significant **architectural upgrade** in decades. The fundamental representation of **money**, how it is **transferred**, **settled**, and **programmed**, is changing not through regulatory mandate but through market-driven **technological adoption**.

The first wave of this transformation, the proliferation of **stablecoins**, has reshaped how value moves across both **permissioned (DLT)** and **permissionless (blockchain)** infrastructure. **Stablecoins**, however, represent only the first phase. A second, more consequential phenomenon is now emerging: **the tokenization of commercial bank deposits**.

Where **stablecoins** are issued by **private entities** and carry **no deposit protection**, **tokenized deposits** are *direct liabilities of regulated commercial banks*. They are the same **regulated money** that has underpinned the **global economy** for centuries, encoded into a programmable, blockchain-native format. **The Bank for International Settlements (BIS)** has identified this distinction as fundamental to the future of the monetary system.

Table 1. Key Indicators of the Tokenized Deposit Market (Q1 2026)

Indicator	Value
Live institutional deployments	17+ across 10+ jurisdictions
BNY Mellon assets under custody	\$52.1 trillion (January 2026)
e-CNY cumulative volume	16.7 trillion CNY
UAE Digital Dirham volume	\$55 billion (mBridge, Q4 2025)
Fnality settlement currencies	5 (GBP, EUR, USD, CAD, JPY)

Sources: BIS (2025); IMF (2026); BNY Mellon press releases; CBUAE (2025); Fnality International prospectus.

2. What are Tokenized Deposits?

1. Legal and Structural Definition

A **tokenized deposit** is a digital representation of a **commercial bank** deposit on a **blockchain** or distributed ledger technology (**DLT**) platform. The definitional distinction from stablecoins is legally significant:

Stablecoins	Tokenized Deposits
– Liability of the issuer (private entity) – Backed by reserve assets held separately – Bearer instrument, freely transferable without restriction – No deposit guarantee scheme coverage – Interest payments prohibited under MiCA and GENIUS Act – Permissionless by design	– Direct liability of a regulated commercial bank – Governed by existing banking law and capital requirements – Transfer-restricted via embedded smart contract controls – Covered by applicable deposit guarantee schemes – Can bear interest on the same basis as conventional deposits – Full AML/KYC compliance embedded at the token level

2.2 The BIS Framework: Three Properties of Sound Digital Money

The **Bank for International Settlements** has identified **three properties** that any form of digital money must preserve to function effectively within the broader monetary system. Tokenized deposits satisfy all three criteria by virtue of their direct integration with the existing two-tier banking structure.

Singleness	Integrity	Elasticity
One unit of tokenized deposit equals one unit of central bank money. No exchange rate friction exists between issuing institutions.	Transactions are final, irrevocable, and legally enforceable. Atomic settlement eliminates double-spend and counterparty risk.	Money supply expands and contracts with economic activity. The credit creation mechanism of commercial banking is preserved.

Stablecoins, by contrast, often compromise on **singleness** (multiple issuers with no par-value guarantee between them) and **elasticity** (full reserve backing removes the credit creation capacity inherent in fractional reserve banking). This structural difference has significant implications for monetary policy transmission.

2.3 Technical Architecture: Two Generations

The technical architecture of **tokenized deposits** has evolved across two generations, each reflecting the regulatory and composability constraints of its era.

Table 2. Architectural Generations of Tokenized Deposits

Attribute	Generation 1	Generation 2
Network type	Private, permissioned consortium	Semi-public or public with permissioned token layer
Access control	Enforced at network level	Embedded in token smart contract
Composability	Limited to closed consortium participants	Open network; DeFi-composable under compliance rules
KYC / AML	Consortium membership controls	Whitelist and transfer restrictions in token logic
Interoperability	Closed consortium only	Cross-protocol and cross-chain capable
Examples	<i>Early JPM Coin, SWIFT pilots (pre-2023)</i>	<i>Kinexys, EURCV, Canton Network (DiSH, Lloyds)</i>

The second generation is now dominant. Rather than restricting the network itself, this architecture embeds **KYC/AML** compliance, **whitelist** controls, and **transfer** restrictions directly into the **token's smart contract**. The underlying network remains open; the asset is controlled. This is the model adopted by **J.P. Morgan's Kinexys**, **Société Générale's EURCV**, and the **Canton Network** deployments of **Lloyds** and **LSEG**.

3. Global Deployments: The State of the Market in 2026

As of Q1 2026, **tokenized deposit** deployments span all major financial jurisdictions. The analysis below documents live and pilot programmes by region, drawing on publicly available institutional disclosures.

Table 3. Global Tokenized Deposit Deployment Matrix (Q1 2026)

Jurisdiction	Initiative	Network / Platform	Status	Primary Use Case
United States	J.P. Morgan Kinexys	<i>Kinexys (private)</i>	Live	FX, treasury automation, programmable payments
United States	Citi Token Services	<i>Private DLT</i>	Live (since 2024)	Cross-border B2B payments (USD/EUR)
United States	BNY Mellon	<i>Private</i>	Live (Jan 2026)	Collateral and margin workflows (\$52.1T AUC)
United States	ICE / NYSE	<i>BNY + Citi rails</i>	Integration (Jan 2026)	Margin clearing outside banking hours
United States	Cari Network (5 regional banks)	<i>TBD</i>	Pilot Q1 2026	Shared interbank settlement
United Kingdom	Lloyds / Archax	<i>Canton Network</i>	Live (Jan 2026)	Out-of-hours repo (atomic DvP)
United Kingdom	LSEG DiSH	<i>Canton Network</i>	Live	Repo cash leg settlement
United Kingdom	GBTD Consortium (7 banks)	<i>TBD</i>	Testing	P2P payments, remortgaging, bond settlement
United Kingdom	Finality International	<i>USC Network</i>	Live	Wholesale interbank (GBP EUR USD CAD JPY)
Germany	CBMT (5 banks)	<i>TBD</i>	Sandbox (Dec 2025)	Treasury automation, machine-to-machine payments
Switzerland	SBA Deposit Token	<i>Public blockchain</i>	PoC complete	Atomic DvP for digital assets
Pan-European	Qivalis (12 banks)	<i>TBD</i>	Development (H2 2026)	MiCA-compliant EUR stablecoin
France	SG-FORGE (EURCV/USDCV)	<i>Ethereum / Morpho</i>	Live (Sep 2025)	Permissioned asset on permissionless DeFi
Japan	DCJPY	<i>DCJPY Network</i>	Live	Multi-bank interoperability
SG / HK	Standard Chartered	<i>Private</i>	Live	Intragroup liquidity (SGD HKD USD CNH)
China	e-CNY (restructured)	<i>State network</i>	Live (Jan 2026)	16.7 trillion CNY cumulative volume
UAE	Digital Dirham (mBridge)	<i>mBridge</i>	Live (Q4 2025)	Cross-border central bank settlement (\$55B)

3.1 United States

J.P. Morgan Kinexys (formerly **Onyx**) is the most **mature tokenized deposit** deployment globally. Operating as a **24/7** institutional **settlement** network, Kinexys supports programmable payments, automated treasury operations, and on-chain FX settlement. In December 2025, BMW Group executed the first fully pre-programmed multi-step FX transaction on the platform, demonstrating the capacity of tokenized deposits to automate complex financial workflows without human intervention.

Citi Token Services (CTS), live since 2024, **enables near-instant cross-border payments in USD and EUR**. Corporate clients interact with their existing fiat accounts while Citi manages the tokenization and settlement layer transparently. As of early 2026, billions of dollars flow through CTS every quarter.

BNY, the world's largest custodian bank with \$52.1 trillion in assets under custody, launched its **tokenized deposit** service in January 2026, initially **targeting collateral and margin workflows**. BNY's entry signals that the custody layer, not just the payment layer, is being rebuilt on tokenized deposit infrastructure. **ICE/NYSE** simultaneously announced integration of tokenized deposits from BNY and Citi across its clearinghouses, enabling clearing members to meet margin obligations outside traditional banking hours.

3.2 United Kingdom

Lloyds Bank executed the first real-money out-of-hours **repo transaction** using **tokenized GBP** on **Canton Network** in January 2026, in partnership with digital securities exchange Archax. A traditional **repo** cannot execute on a Saturday night because the cash leg requires settlement through systems that operate only during banking hours; **tokenized GBP** enabled **atomic Delivery versus Payment (DvP)** settlement outside those hours for the first time.

LSEG's Digital Settlement House (DiSH) uses tokenized deposits as the cash leg for **repo financing**, representing one of the most significant institutional endorsements of the technology's readiness. The **Great British Tokenized Deposit (GBTD) consortium**, comprising **Barclays, HSBC, Lloyds, NatWest, Nationwide, Monzo, and Santander**, is testing peer-to-peer payments, remortgaging, and bond settlement. **Finality International** (17 global financial institutions) operates across five currencies outside RTGS hours.

3.3 Continental Europe

Germany: The **CBMT consortium (Deutsche Bank, Commerzbank, DZ Bank, Helaba, LBBW)** entered sandbox phase in December 2025 with **Bosch, Siemens, and BMW** as corporate pilots, focusing on treasury automation and machine-to-machine (M2M) payments.

Switzerland: The **SBA Deposit Token** proof-of-concept, executed by **PostFinance, Sygnum Bank, and UBS**, tested interbank payments and atomic DvP settlement for digital assets on public blockchain infrastructure.

Pan-European: Qivalis, a joint venture of eleven European banks (BNP Paribas, UniCredit, ING, Santander, SEB, KBC, and others), is developing a **MiCA-compliant euro stablecoin** with a launch target of H2 2026.

France: The SG-FORGE Model: Among all European deployments, **Société Générale-FORGE** has articulated the most strategically significant architecture. In September 2025, SG-FORGE deployed **EURCV** and **USDCV**, its **MiCA-compliant stablecoins**, directly into **Morpho's permissionless lending infrastructure**.¹ **Morpho** is a **fully permissionless lending protocol**; **SG-FORGE** embedded **compliance** controls into the **token** itself. The result: any eligible wallet can **borrow EURCV** from a **Morpho vault** while the compliance checkpoint moves from market access to **FIAT** redemption. This *compliant asset on open infrastructure* model is likely to become the dominant template for institutional DeFi participation.

3.4 Asia-Pacific

Japan: DCJPY, developed by **DeCurret DCP**, represents the most interoperable domestic deployment in Asia, based on a common token standard enabling mutual **fungibility** between **deposits** issued by different participating **banks**.

Singapore / Hong Kong: Standard Chartered has deployed **tokenized deposits** across **SGD, HKD, USD, and CNH**, enabling real-time intragroup liquidity management across geographically dispersed entities.

China: The **e-CNY** underwent a structural transition in January 2026, moving from a pure central **bank** liability to a **commercial bank tokenized deposit** structure. Cumulative transaction volume reached **16.7 trillion yuan**.

3.5 Middle East

The **UAE Digital Dirham**, launched in Q4 2025 on the **mBridge** platform, enables direct **cross-border central bank** settlement between the **UAE, China, Hong Kong, and Thailand**, with **\$55 billion** in transaction volume. **Zand Bank and RAKBANK** have issued **AED-backed stablecoins** on public blockchain infrastructure, marking the first **UAE commercial bank** deployments of dirham-denominated digital money on **open rails**.

Table 4. Key Deployment Milestones (2024–2026)

Date	Milestone	Institution
2024	Citi Token Services goes live: cross-border USD/EUR settlements	Citi
Sep 2025	SG-FORGE deploys EURCV/USDCV on Morpho permissionless protocol	SG-FORGE
Dec 2025	CBMT enters sandbox; BMW executes first pre-programmed FX on Kinexys	JPM / CBMT
Dec 2025	e-CNY restructured as commercial bank tokenized deposits	China PBoC
Q4 2025	UAE Digital Dirham launches on mBridge: \$55B transaction volume	CBUAE
Jan 2026	Lloyds executes first out-of-hours repo using tokenized GBP	Lloyds / Archax
Jan 2026	BNY launches tokenized deposit service (\$52.1T AUC)	BNY
Jan 2026	ICE/NYSE integrates tokenized deposits across clearinghouses	ICE / NYSE
H2 2026	Qivalis MiCA-compliant EUR token: planned launch	Qivalis (11 banks)

4. Regulatory Landscape

Table 5. Regulatory Framework by Jurisdiction

Jurisdiction	Framework	TD Classification	Interest Allowed	Outlook
United States	GENIUS Act (2025)	Explicitly excluded from stablecoin definition	Yes	Clear regulatory path; favorable for banks
European Union	MiCA	Ambiguous; potential e-money classification	Pending clarity	Clarification expected Q4 2026
United Kingdom	FCA Innovation Hub	Supervised case-by-case approval	Yes	Progressive; sandbox-first approach
Singapore	MAS Guidelines	Case-by-case approval	Yes	Innovation-led; pragmatic
Japan	FSA Framework	Bank deposit (regulated)	Yes	Clear framework; DCJPY deployed
China	PBoC Directive	State-managed commercial bank deposit	Yes	Centrally directed; e-CNY restructured
UAE	CBUAE Regulation	Central bank governed	Yes	Proactive; mBridge leadership

4.1 The United States: Clarity Through Exclusion

The **GENIUS Act** explicitly excludes bank deposits from the definition of payment stablecoins, preserving the **OCC/Federal Reserve/FDIC** framework for tokenized deposits. This regulatory clarity has accelerated **institutional deployment** by eliminating the primary source of uncertainty that had delayed programmes in earlier years.

4.2 The European Union: Ambiguity as Competitive Disadvantage

Under MiCA, **e-money tokens cannot pay interest**. If **tokenized deposits** are classified as **e-money tokens**, their **yield-bearing capability**, a primary structural advantage over **stablecoins**, **would be eliminated**. The EU's regulatory ambiguity creates a **competitive asymmetry**: **institutions** in the **United States and Asia-Pacific** can deploy with certainty while **European banks** await clarification expected by Q4 2026.

4.3 Asia-Pacific: Pragmatic Deployment

Singapore, Hong Kong, Japan, and China have each adopted use-case driven **regulatory approaches**, enabling **deployments** such as **DCJPY**, Standard Chartered's network, and the **e-CNY** restructuring to proceed without waiting for comprehensive legislative frameworks. The pragmatism of Asia-Pacific regulators stands in contrast to the EU's more prescriptive approach.

5. Strategic Implications

1. The Three-Layer Digital Money Architecture

Evidence across deployments suggests that **future digital money** systems will operate as a **three-layer architecture**, with each instrument occupying a distinct functional role:

Layer	Instrument	Primary Function	Key Networks
Layer 3 (Interbank)	Wholesale CBDCs	Central bank settlement; interbank clearing	<i>mBridge, Project Dunbar, Fnlity</i>
Layer 2 (Institutional)	Tokenized Deposits	Settlement, treasury management, collateral workflows	<i>Kinexys, CTS, Canton, DCJPY, DiSH</i>
Layer 1 (Retail / DeFi)	Stablecoins	DeFi protocols, retail payments, cross-border remittance	<i>USDT, USDC, EURCV, Qivalis EUR</i>

5.2 Banks as the Foundation Layer

Stablecoins and **tokenized deposits** are not competitors but **complementary** instruments. Banks issuing **tokenized deposits** are simultaneously the **primary** providers of **stablecoin reserve** custody, **FIAT on/off ramps**, and settlement infrastructure. The institutions that control the **tokenized deposit** layer therefore capture the highest-margin segment of the digital money value chain regardless of whether stablecoins or tokenized deposits dominate any particular use case.

5.3 The Infrastructure Opportunity

The **SG-FORGE / Morpho** model demonstrates that **regulated institutions** will **leverage** existing **DeFi infrastructure** rather than building proprietary lending protocols. Competitive advantage will accrue to **DeFi protocols** offering *isolated, configurable markets with institutional-grade risk controls*. For **banks**, the strategic imperative is to **move** from **pilot** to **production**. For infrastructure providers, it is to build for institutional composability. For **regulators**, it is to **provide framework clarity** before regulatory arbitrage shifts activity permanently to more permissive jurisdictions.

Table 6. Strategic Positioning Matrix

Actor	Strategic Opportunity	Risk if Inactive
Global banks	Issue tokenized deposits; own custody and settlement layer	Disintermediated by faster-moving fintech and regional banks
Regional banks	Join consortia (Cari, GBTD) to achieve network scale	Lose institutional and corporate clients to larger banks
DeFi protocols	Build institutional isolated markets (Morpho model)	Miss the largest inflow of regulated capital into DeFi
Custodians	Rebuild collateral layer on tokenized deposit rails	Lose margin and collateral business to clearinghouses
Regulators	Provide classification clarity (especially EU)	Regulatory arbitrage: activity migrates to US and Asia

6. Conclusion

The **tokenization of commercial bank deposits** represents the **second** and more consequential **wave of digital money transformation**. Unlike **stablecoins**, which introduced a new category of money issuer, **tokenized deposits** work *within* the existing **two-tier banking system**, encoding established regulatory protections into a programmable format.

The deployments documented in this paper, **J.P. Morgan's Kinexys**, **Lloyds' out-of-hours repo**, **DCJPY's** multi-bank standard, **SG-FORGE's permissioned-on-permissionless** architecture, **BNY's** custody integration, and the **e-CNY** restructuring, demonstrate that **tokenized deposits** are no longer theoretical. They are live, processing real transactions, and embedding themselves into the settlement layer of core financial infrastructure.

The question is no longer whether **tokenized deposits** will be adopted. It is which institutions will define the **infrastructure** layer, and which will remain dependent on those that do.

Notes

1. SG-FORGE deployed EURCV and USDCV on Morpho Protocol in September 2025, the first instance of a MiCA-compliant institutional stablecoin operating within a fully permissionless DeFi lending market. The token carries embedded compliance controls (whitelist, KYC gate at fiat redemption); the underlying Morpho protocol remains permissionless. Source: Morpho Protocol documentation; SG-FORGE Regulatory Disclosure (September 2025).

References

- 1 **Bank for International Settlements (BIS)** (2023). *The tokenisation continuum*. BIS Bulletin No.†72. Basel: BIS.
<https://www.bis.org/publ/bisbull72.htm>
- 2 **Bank for International Settlements (BIS)** (2022). *The future monetary system*. BIS Annual Economic Report, Ch. III.
<https://www.bis.org/publ/arpdf/ar2022e3.htm>
- 3 **Deloitte** (2026). *Payments 2026: Stablecoin adoption among financial institutions*. Deloitte Center for Financial Services, Global Outlook Report.
- 4 **J.P. Morgan** (2025). *Kinexys Digital Payments: Platform Overview and Use Cases*. J.P. Morgan Institutional Solutions.
<https://www.jpmorgan.com/kinexys>
- 5 **Societe Generale — FORGE** (2025). *EURCV and USDCV: Deployment on Morpho Protocol*. SG-FORGE Regulatory Disclosure. <https://sg-forge.com>
- 6 **European Parliament and Council of the EU** (2023). *Regulation (EU) 2023/1114 on Markets in Crypto-Assets (MiCA)*. Official Journal of the EU, L 150/40.
<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32023R1114>
- 7 **United States Congress** (2025). *GENIUS Act: Guiding and Establishing National Innovation for US Stablecoins*. S.394, 119th Congress.
- 8 **Fnality International** (2025). *The Utility Settlement Coin: Wholesale Payment Infrastructure*. Fnality White Paper. London. <https://fnality.org>
- 9 **Digital Asset Holdings** (2025). *Canton Network: Privacy-Enabled Institutional DLT*. Canton Network Technical Overview. <https://canton.network>
- 10 **DeCurret DCP Inc.** (2025). *DCJPY: Digital Currency Joint Research Initiative*. Digital Currency Forum Progress Report. Tokyo. <https://dcjpy.com>
- 11 **Bank of England / FCA** (2025). *Digital Securities Sandbox and Tokenised Deposit Infrastructure*. FCA Innovation Hub Progress Report. London. <https://www.bankofengland.co.uk>
- 12 **Citigroup** (2024). *Citi Token Services: Real-Time Cross-Border Payments*. Citi Treasury and Trade Solutions Product Brief. <https://www.citigroup.com>
- 13 **BNY Mellon** (2026). *Tokenized Deposit Service: Collateral and Margin Workflows*. BNY Digital Assets Product Announcement. <https://www.bnymellon.com>
- 14 **Adrian, Tobias (IMF)** (2026). *Tokenized Finance*. IMF Note 2026/001. International Monetary Fund, Washington DC.
<https://www.imf.org/en/Publications/Notes>
- [14b] **International Monetary Fund (IMF)** (2023). *Digital Money and the Future of the Monetary System*. IMF Staff Discussion Note SDN/2023/001. Washington D.C. <https://www.imf.org/en/Publications/SDN>
- 15 **Morpho Labs** (2025). *Morpho Protocol: Isolated Markets and Institutional Risk Controls*. Morpho Protocol Technical Documentation. <https://morpho.org/documentation>
- 16 **Central Bank of the UAE (CBUAE)** (2025). *Digital Dirham and mBridge Cross-Border CBDC Platform*. CBUAE Financial Infrastructure Report. Abu Dhabi. <https://www.centralbank.ae>

- 17 **People's Bank of China (PBoC)** (2025). *e-CNY Restructuring as Commercial Bank Tokenized Deposits*. PBoC Digital Currency Research Institute. Beijing.
- 18 **Swiss Banking Association (SBA)** (2024). *Swiss Deposit Token: Proof-of-Concept on Public Blockchain*. SBA Digital Asset Working Group Technical Report. Berne. <https://www.swissbanking.ch>